Managerial Economics

Lecture 7 Firm Behavior in Different Market Structures

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RMBS

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RENMIN BUSINESS SCHOOL

Roadmap

- Perfect Competition
- Imperfect Competition
  - Monopoly
  - Monopolistic Competition
  - Oligopoly
- Market Concentration Index

Perfect Competition

Market Equilibrium in Perfect Competition

- **Perfect competition(完全竞争):** firms are price takers
  - Large number of relatively smaller buyers and sellers
  - Standardized or homogenous products, e.g. rice, wheat, water
  - Free entry and exit
  - Perfect information on product quality and price

- In perfect competition, *industrial supply* ***equals*** *industrial demand* determines the market equilibrium price and quantity

Pricing and output decisions in perfect competition

- **Marginal revenue/Marginal cost approach to determine optimal output:**
  - produce a level of output at which the additional revenue received from the last unit is equal to the additional cost of producing that unit

$$MR(Q) = MC(Q)$$

  - In a perfectly competitive market, every extra product you sell, you earn marginal revenue:

$$MR(Q) = P$$

  - Profit maximization means the optimal output level satisfies that $P = MC(Q^*)$

In perfectly competitive market, firms are price takers, they can only adjust their output levels to maximize their profits.

Pricing and output decisions in perfect competition

Profit maximization implies that each firm produces an output where Price = Marginal Cost (**P = MC**).

- To produce more than this quantity implies that $P < MC$, which is not the most profitable decision.
- To produce less than where P=MC, implies that $P > MC$, and the firm could increase profits by expanding output.

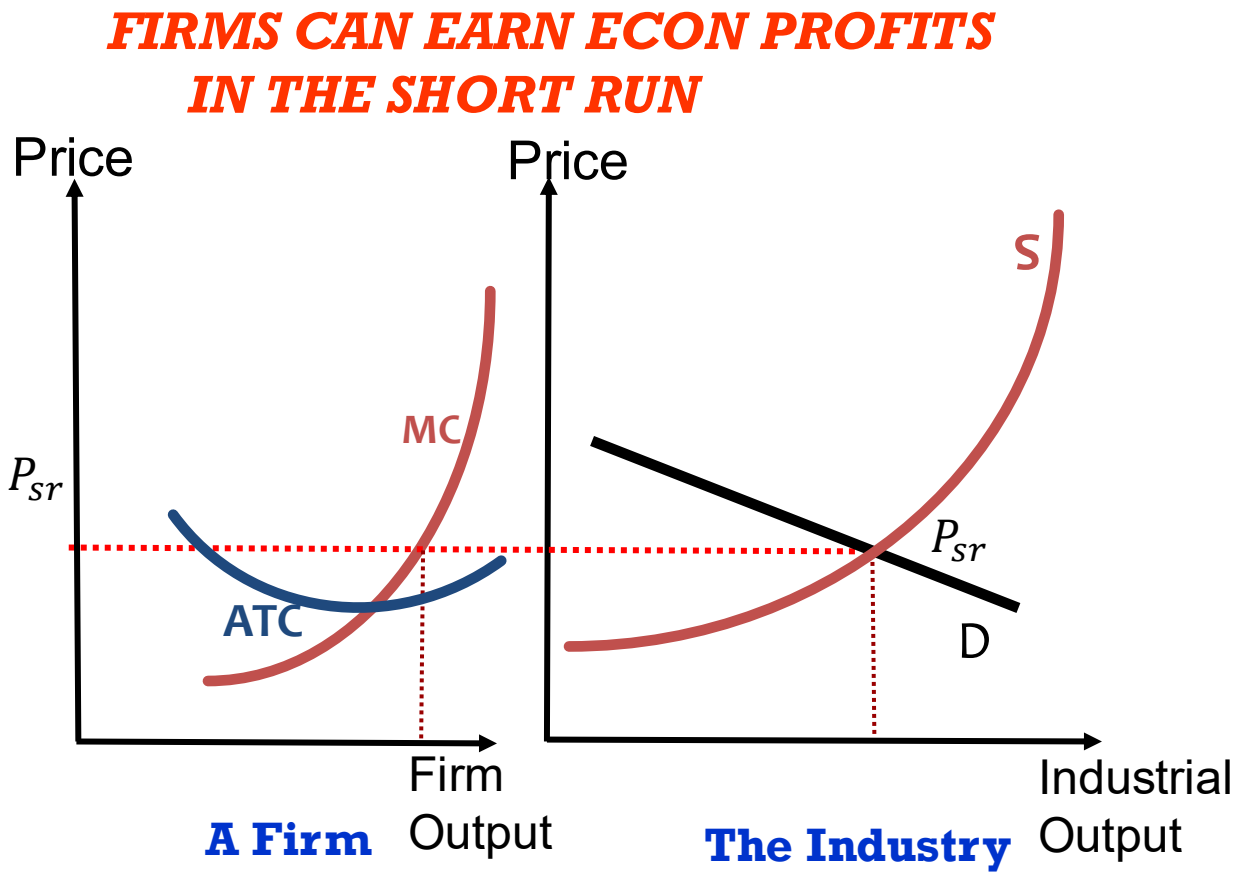
- In the short run, a competitive firm may **earn economic profits**.
- In the long run, **new market entrants** push down price to the minimum point of the average cost curve, so that economic profits are zero.

A Competitive Market in the Short Run

- Equilibrium for the industry if: *Demand equals Supply*
- If $MR=P=MC$ for each firm, then each firm is maximizing profits; Firms are in equilibrium.
- In this example, the firm is earning economic profits as

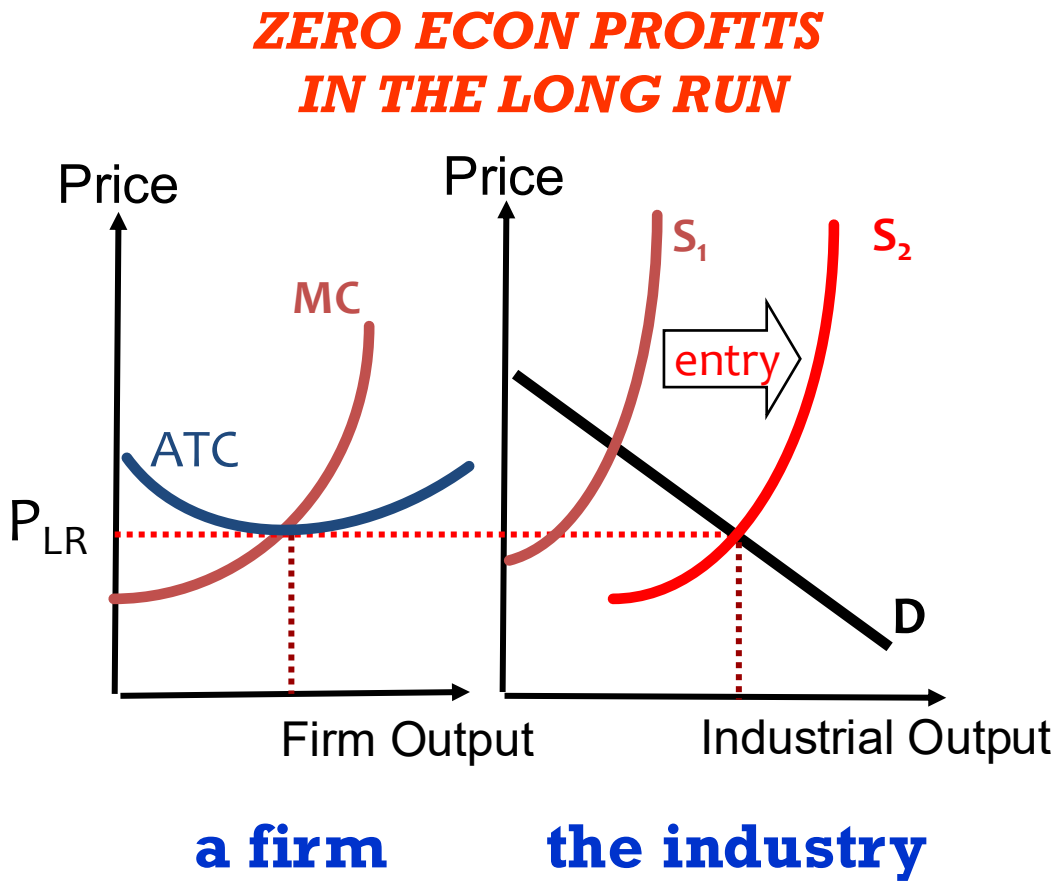
$$P_{sr} > ATC$$

  - *Cost leadership strategy*
  - *Product differentiation strategy*



Long-run Economic Profits in the Long Run

- Industries which have economic profits attract new entrants
- Shift the S_1 curve to S_2
- Price covers all cost, so in the LR, P=AC which means that Economic Profits are zero.



A Competitive Market in the Long Run

- **Economic profit equals zero in the long run**
- *Does it make any sense for a firm to stay in the perfectly competitive market?*

  - Yes
  - Economic profit =0 → Revenue - opportunity cost （explicit cost + implicit cost） =0
  - Revenue covers explicit cost & opportunity cost
  - Both managers & workers get paid; interests and machine depreciation are covered!

Practice: Market Equilibrium in Perfect Competition

Assume that the market is perfectly competitive:

- For the industry:
  - Supply Function: $Q_S = 3000 + 200\ P$
  - Demand Function: $Q_D = 13500 - 500\ P$

- For the firm:
  - Fixed Cost: FC = 50
  - Marginal Cost: MC = 3Q

Calculate optimal **price** and **output** for this firm.

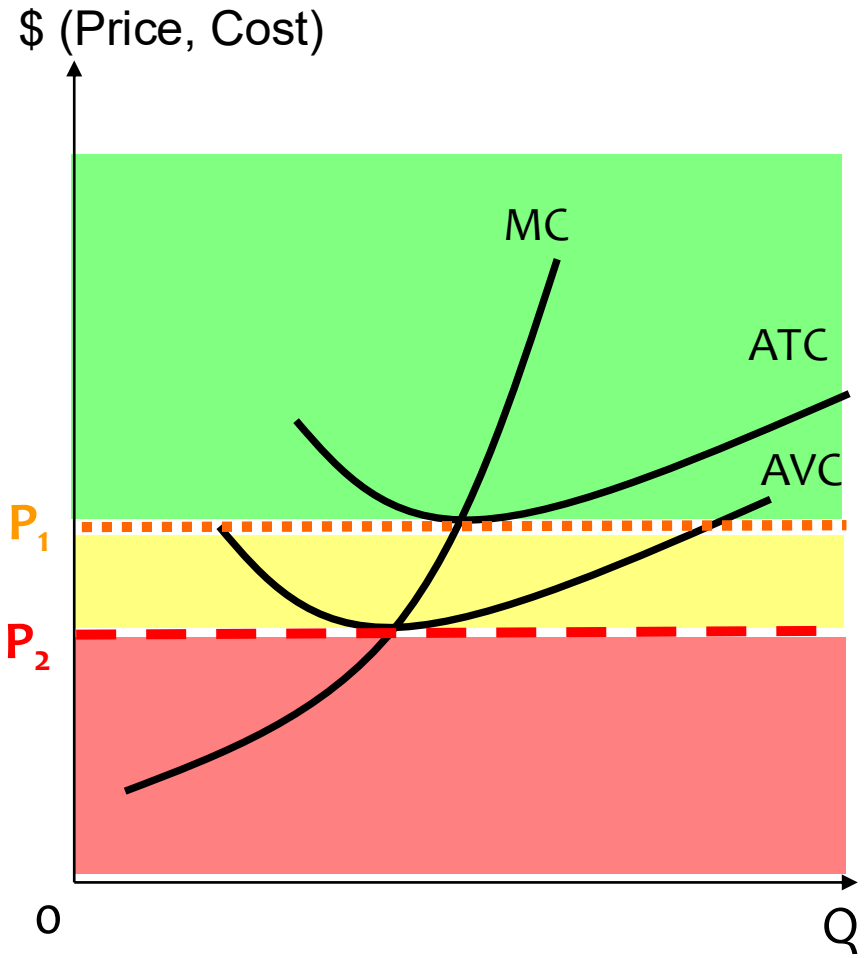
Breakeven and Shutdown Points

- **Breakeven point (收支平衡点, P1):** the point where price is equal to average total cost

$$P_1 = ATC$$

- **Shutdown point (停业点, P2)**: the lowest price at which the firm would still produce
  - At the shutdown point, the price is equal to the **minimum point** on the AVC
  - If the price falls below the shutdown point, revenues fail to cover the fixed costs and the variable costs. The firm would be better off if it shut down and just paid its fixed costs

- *P*>*P1*, positive profit
- P2<P<P1, cover part of fixed cost
- P<P2, close down



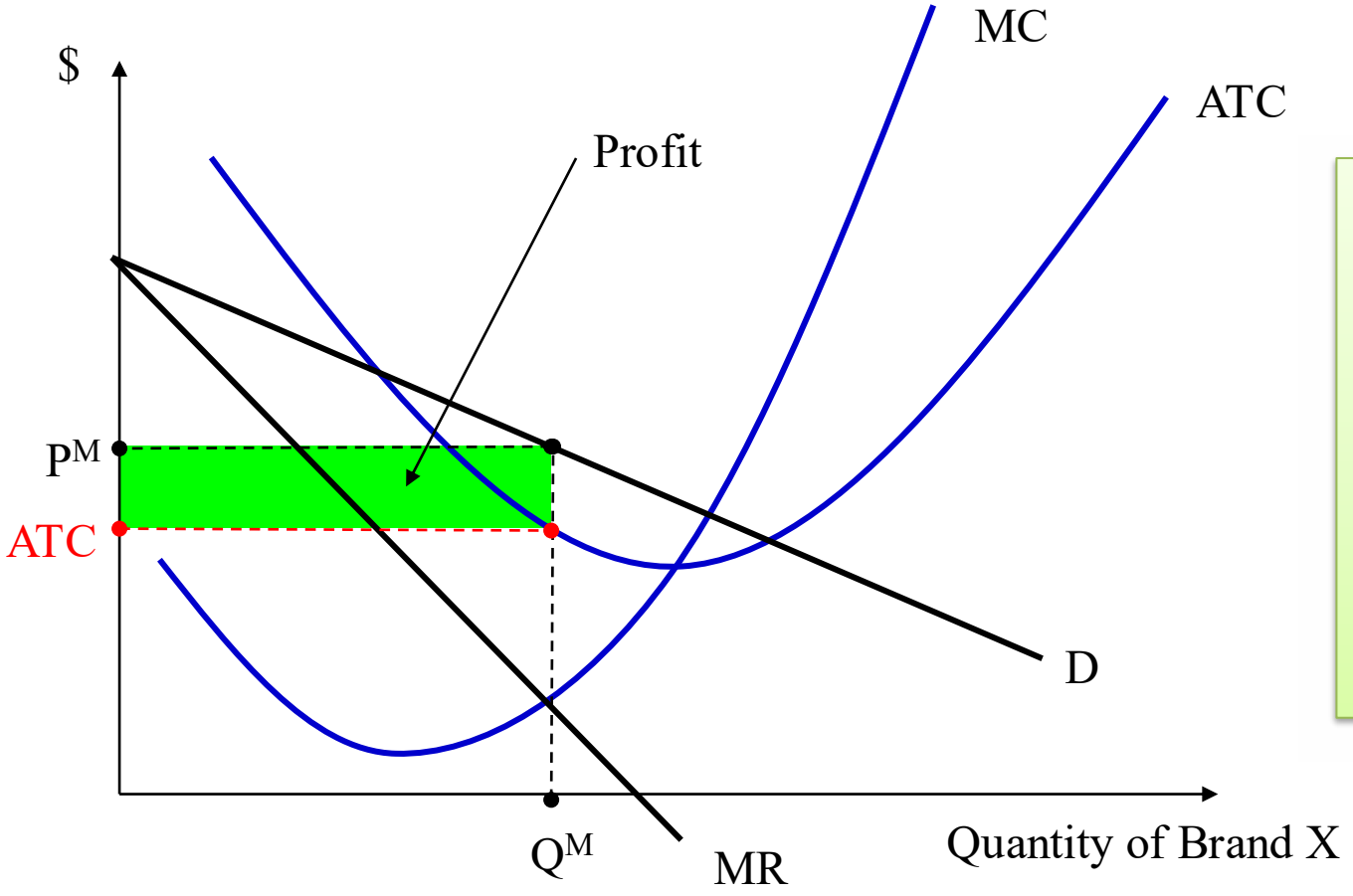
Imperfect Competition

Monopolistic Competition

Monopolistic Competition: Characteristics

- **Monopolistic competition（垄断竞争）**
  - many firms
  - Free entry and exit
  - product differentiation: can set price at a level higher than the price established by perfect competition
  - possess a certain degree of market power over pricing, i.e. not a pure "price taker"
  - again, use "golden rule" MR = MC to maximize profit

Short-run Monopolistic Competition



By setting price over the average total cost, monopolistic competitive firm can earn positive profits in the short run.

Monopolistic Competition in the Long Run

- Economic profit invites entry of new firms → shifts the supply curve to the right → puts downward pressure on price and reduces profits

- Economic loss causes exit of firms → shifts the supply curve to the left → puts upward pressure on price and increases profits

Product Differentiation in Monopolistic Competition

- The core difference between monopolistic competition and perfect competition lies in product differentiation.
  - When consumers perceive a product as **difficult to be substituted** by products from other firms in the same industry, the **demand elasticity** of consumers tends to be **low**.
  - The **lower the demand elasticity**, **the greater the profit margin** for the company.
- Monopolistically competitive firms continuously persuade consumers that their products are superior to those of their competitors.
  - For example, in the bottled water industry, various types such as purified water, mineral water, mineralized water, natural water, weak alkaline water, and small-molecule mineral water……
- Product differentiation strategies include:

① **Comparative Advertising (比较型广告)**: An advertising strategy that attempts to enhance demand by distinguishing a product from competing brands.

  - Celebrity Brand Endorsement

② **Developing New Products**

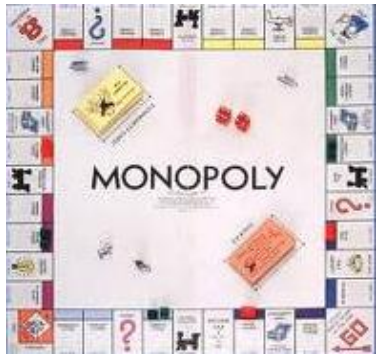
  - New Improved Products
  - Completely New Products
  - Niche Market (利基市场): A market space occupied by a small group of customers within a larger segmented market, who share similar interests or needs.

Imperfect Competition

Monopoly

Monopoly and Dominant Firms (垄断与支配性企业)

- A monopoly（垄断） is a firm that is the only seller of a product without close substitutes.
- The key difference between monopoly and perfect competition :
  - A monopoly firm has **market power**（垄断势力；市场操作力）, the ability to influence the market price of the product it sells.
  - A competitive firm has no market power.
- Just as purely competitive market structures are rare, so too pure monopoly markets are also rare.
- We usually consider a dominant firm（主导企业） rather than looking for a *pure monopoly*.



Monopoly and Dominant Firms (垄断与支配性企业)



"*Monopoly*" conjures images of huge profits, great wealth, and absolute power, labeled ***robber barons***（强盗大亨）.

- But some monopolies are strongly regulated by the government, and may have low rates of return on invested capital, e.g. *Beijing Subway*
- Firms may need to pay costs to obtain monopoly power

Why Monopolists Arise?

The main cause of monopolies is **barriers to entry（进入壁垒）** – other firms cannot enter the market.

- **Legal restrictions** -- copyrights & patents.
- **Control of critical resources** creates market power – gold mine, crude oil
- **Government-authorized franchises（特许权经营）**, such as those provided to cable TV companies.
- **Brand loyalty （品牌忠诚度）and extensive advertising** makes entry highly expensive, e.g. Prada
- **Increasing returns in network-based businesses（基于网络的规模报酬递增型行业）** - compatibilities increase market penetration, e.g. WeChat

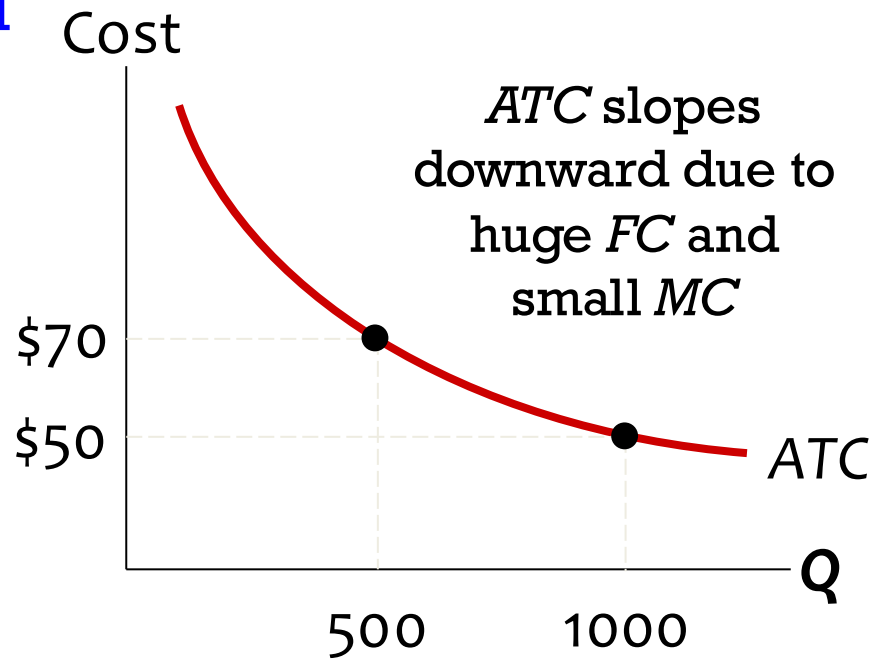
Why Monopolies Arise?

- **Natural monopoly（自然垄断）**: a single firm can produce for the entire market at a lower cost than other firms due to economies of scale

Example: 1000 households need tap water

ATC is lower if one firm serving all 1000 households than if two firms each serving 500 households.

Local water supply



A Monopoly's Revenue

The table shows the market demand for a monopoly.

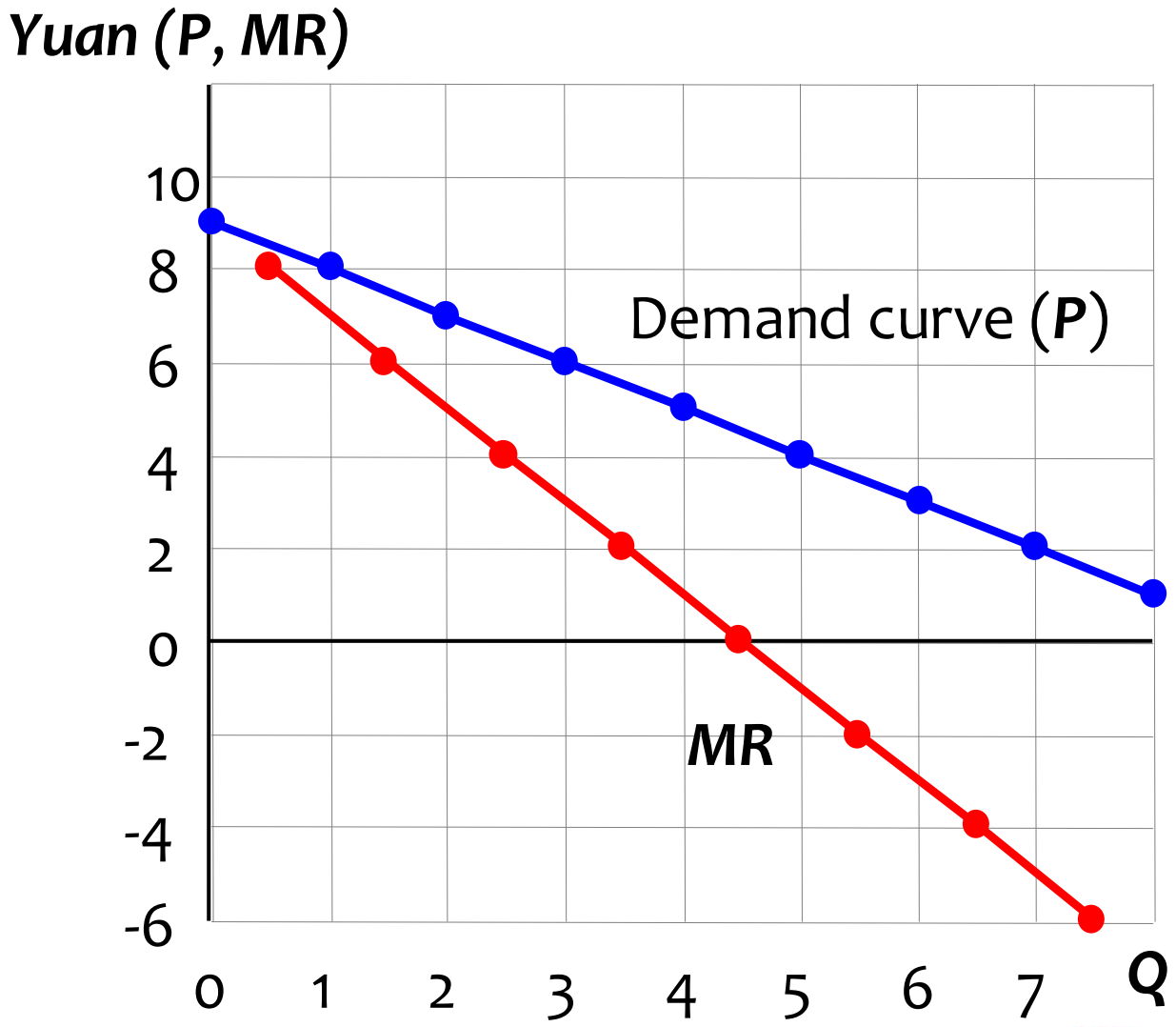
Fill the missing spaces of the table .

What is the relation between ***P*** and ***AR***? Between ***P*** and ***MR***?

Q	P	TR	AR	MR
0	9		n.a.	
1	8			
2	7			
3	6			
4	5			
5	4			
6	3			

The Marginal Revenue Curve for A Monopoly

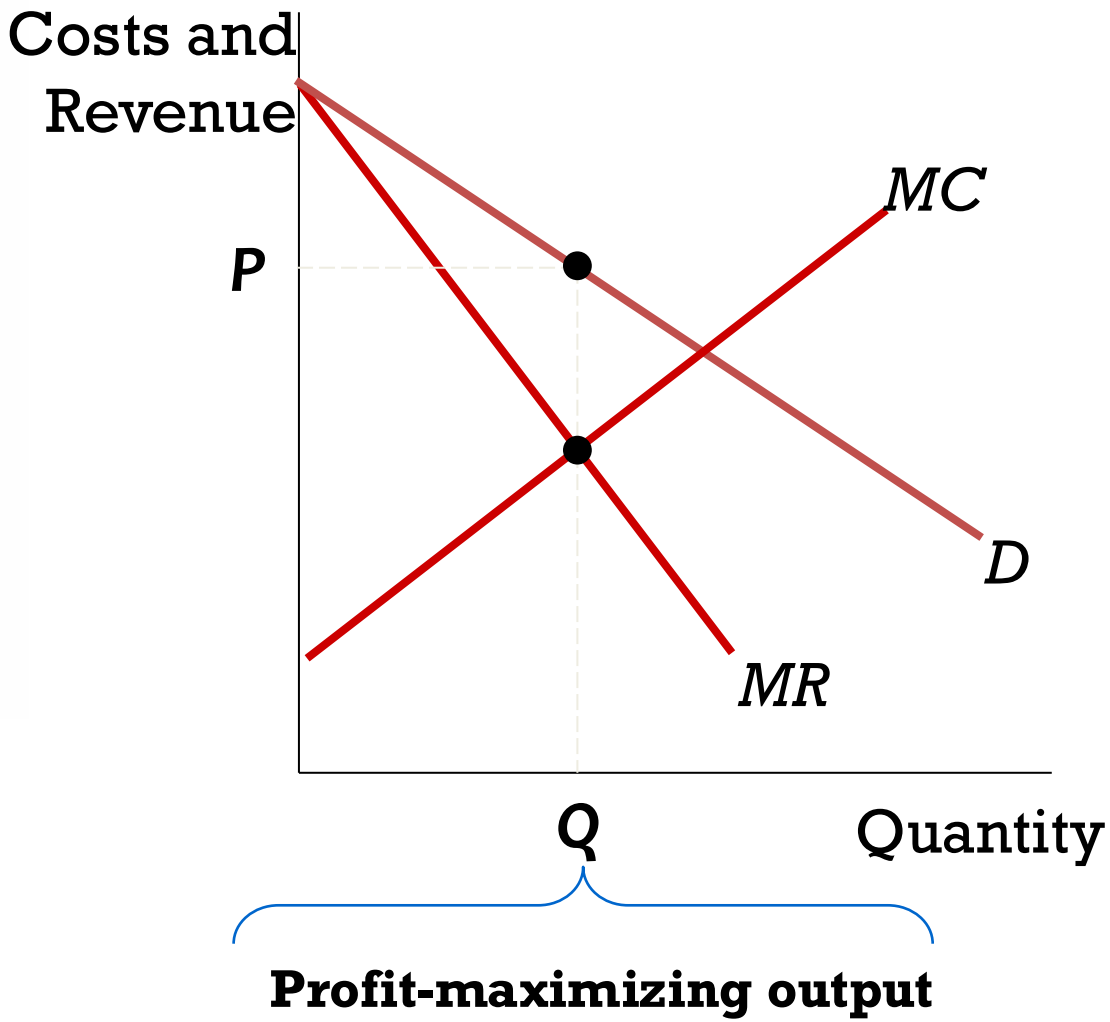
Q	P	MR
0	9	
1	8	8
2	7	6
3	6	4
4	5	2
5	4	0
6	3	−2



Profit-Maximization of A Monopoly

1. The profit-maximizing Q is where $MR = MC$.

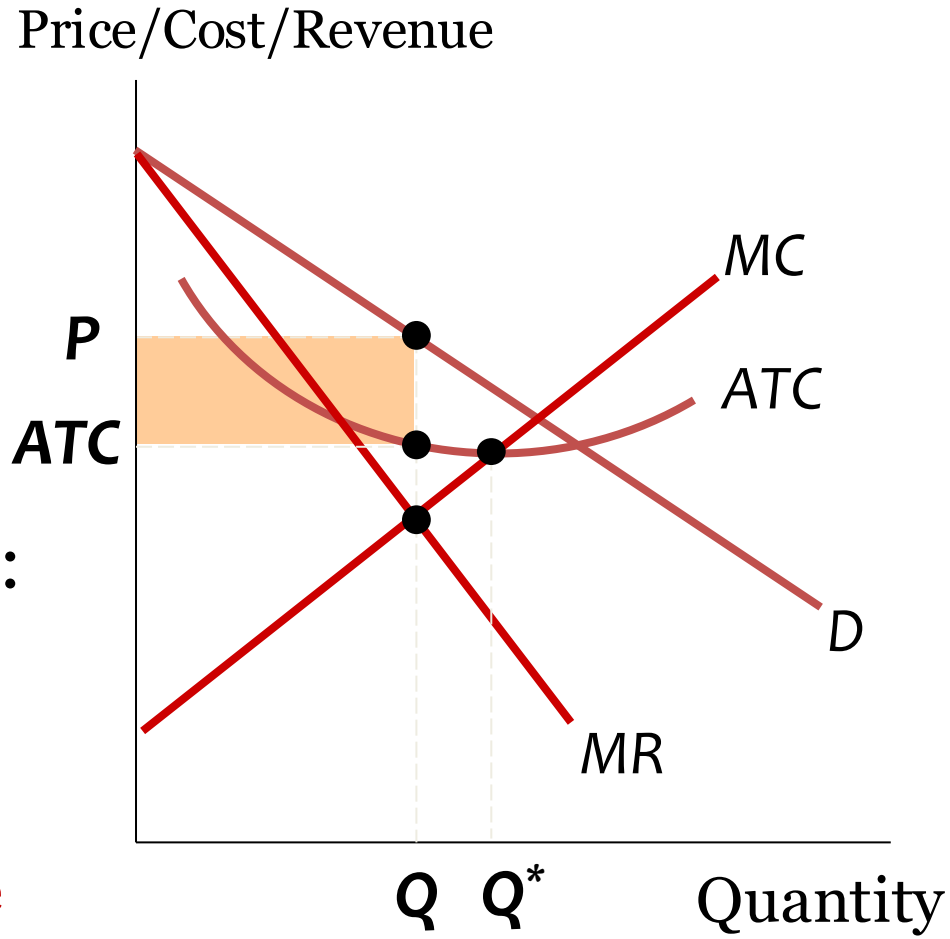
2. Find P from the demand curve at this Q.



The Monopolist's Profit

- The monopolist's profit equals

$$\underbrace{(P - ATC)}_{\textit{profit margin}} \times Q$$



- Two (possible) important features:
  1. **P>MC :** The monopolist generate positive profits
  2. **Q<Q*** (where ATC is the lowest): The monopolist may not produce in the most efficient way

Real-life Monopolists

- The largest taxi com
  without taxi

DiDi

- The largest retailer, without
  inventory

Taobao

- The largest user base, without
  charges

360

- The largest search engine,
  without …

Baidu

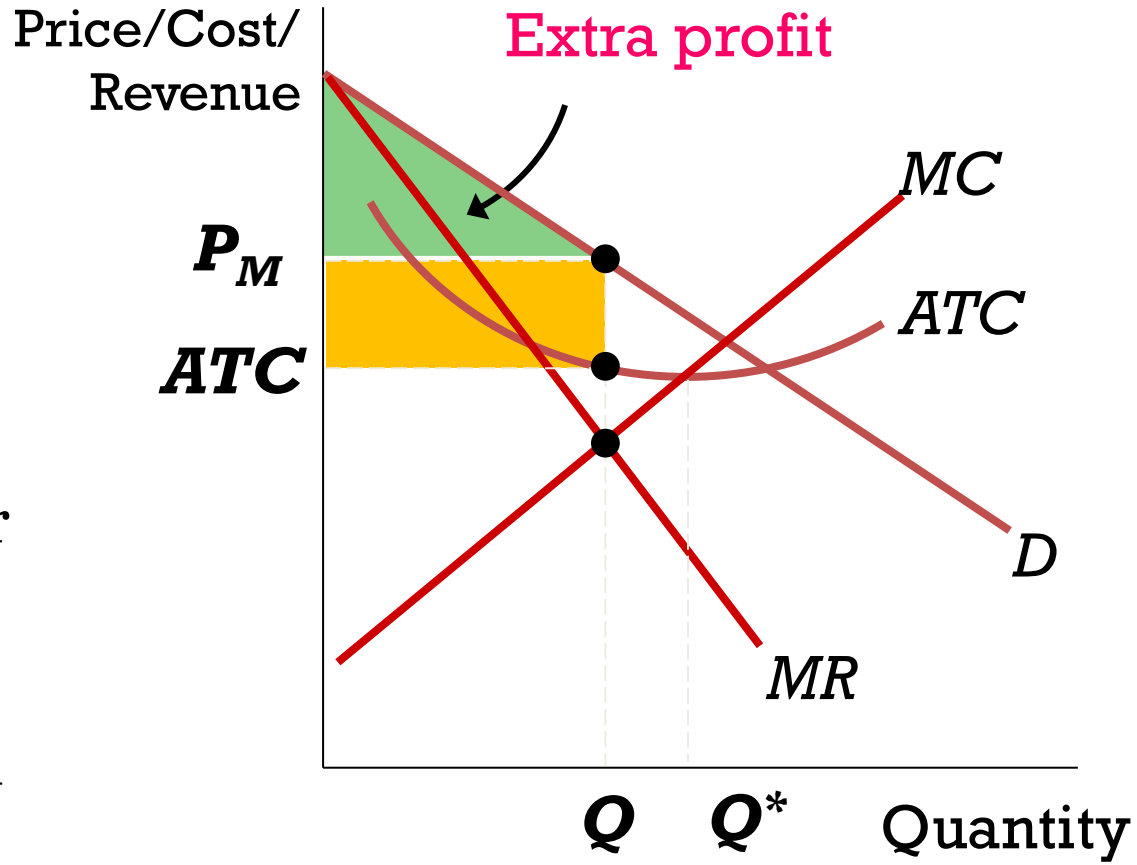
- The hottest media, without original
  reports

WeChat

The Monopolist's Profit under Perfect Price Discrimination (价格歧视)

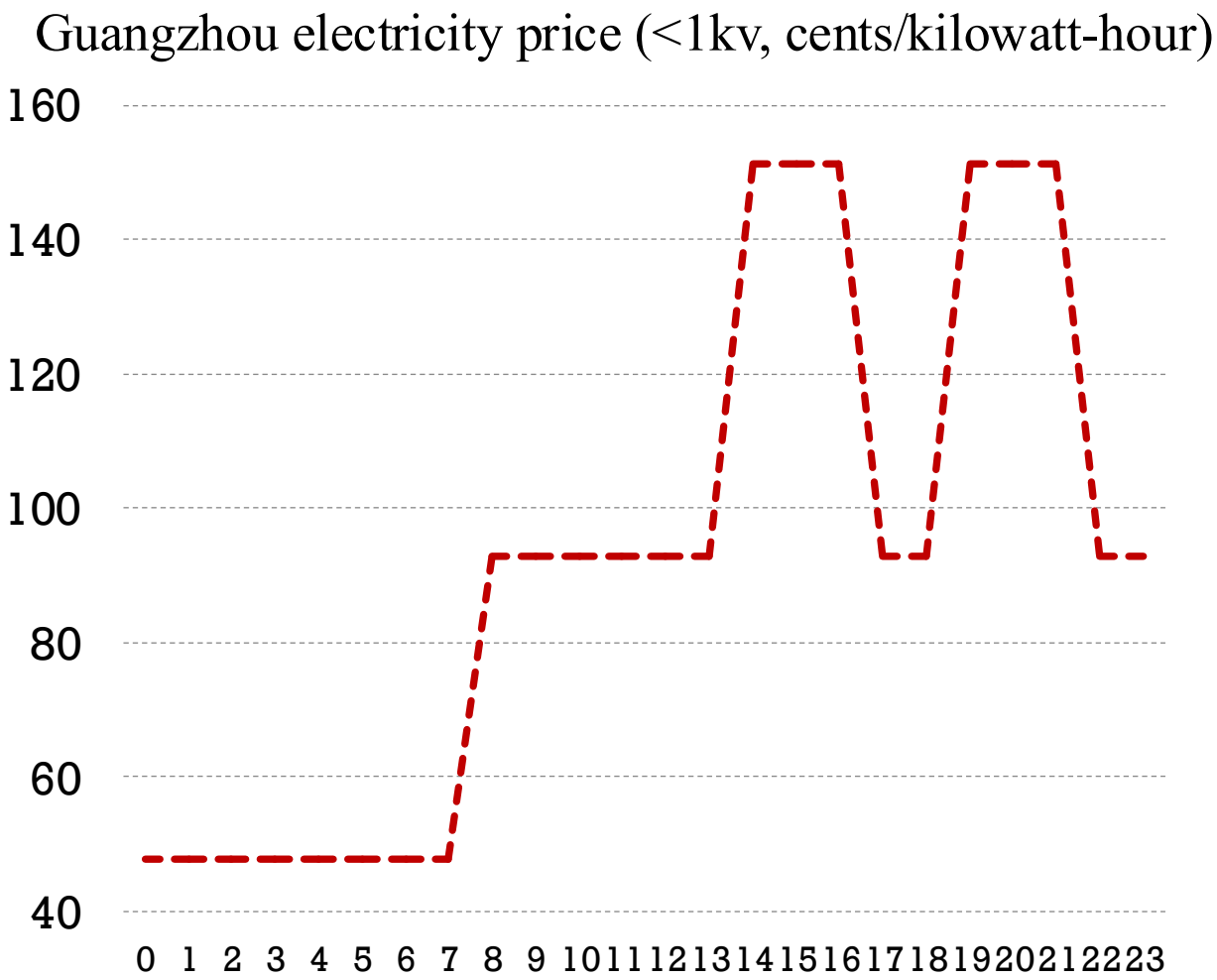
- If the monopolist charges the same price (P_M) to all buyers, it can only get the **orange profit**.

- But if the monopolist charges prices above P_M for customers with higher willingness to pay (WTP), its profits will increase with the amount of the **green triangle.**



Examples of Price Discrimination

- **Movie tickets / Gym memership**
  Discounts for students, children, and people who can attend during weekday afternoons. They are all more likely to have lower WTP than people who pay full price on Friday night.
- **Airline prices**
  Discounts for early bookings help distinguish business travelers, who usually have more urgent arrangements & higher WTP, from more price-sensitive leisure travelers who usually book much in advance.
- **Peak Load Pricing (错峰定价)**



Social Welfare Cost of Monopoly

- Governments in all countries have strict regulations over the pricing of monopolistic firms. Why? …
- In most cases, in the monopoly equilibrium
  1. ***P > MR = MC***, the money buyers need to spend on an additional unit (***P***) exceeds the cost of the resources needed to produce that unit (*MC*).
  2. ***The monopoly Q is too low*** – more consumers could have consumed this product or service.
- Therefore market ("invisible hands"看不见的手) itself can not reach its most efficient status: market failure（市场失灵）.
- When market malfunctions, government must do something!

Regulated Monopolies（受管制的垄断）

- Electric power companies
- Natural gas companies
- Communication companies (telephone, cable, radio, TV, etc.)
- Municipal water companies
- Public transportation companies

Solutions to the Problem of Natural Monopolies

- **Prevent entry + Set P = MC + Subsidize.**
  - Subsidies require some form of taxation, which will tend to distort workers' incentives.
- **Nationalize（国有化）+ Prevent entry + Set price typically low**
  - Governments find that changing its price is a highly political event
- **Regulate, prevent entry + set P = AC**
  - Common in US for electricity, water
- **Franchise（特许权经营） through a bidding war + P = AC**
  - Cable T.V.
- **Doing nothing (laissez-faire)（放任自流）**
  - Policies all have distortions, so the best policy may be no policy
  - Natural monopoly is more cost-effective than any separate operation!

Microsoft Anti-trust Case

- In the 1990s，U.S. federal regulators sued Microsoft, which was at that time the world's leading software company. The Federal Trade Commission launched an investigation as a response to the rising market share of the company in the personal computer market.
- The biggest argument made by Microsoft's defenders was that **antitrust laws stifle the success of domestic firms on a global level**, hence making them less competitive.

Imperfect Competition

Oligopoly

Oligopoly

- **Oligopoly (寡头竞争)**: a market structure in which only a few sellers offer similar or identical products.
- **Strategic actions and reactions in oligopoly**
  - A firm's decisions about *P* or *Q* can affect other firms and cause them to react.
  - The firm will consider these reactions and consequences when making decisions.
- **Game theory（博弈论）**: the study of how people, firms, and countries behave in strategic situations.

Actions & Reactions

- **Alibaba + Weibo**
  - Equity share: 18% in Weibo in 2013 → 30%







- **Tencent + JD**
  - Equity share: 15% in JD in 2014 → 17.8%







EXAMPLE: Duopoly in Smalltown

P	*Q*
$0	140
5	130
10	120
15	110
20	100
25	90
30	80
35	70
40	60
45	50

- Smalltown has 140 residents
- Both provide mobile phone service with unlimited anytime minutes and free phone
- Smalltown's demand schedule
- Two firms: China Mobile, China Unicom (**duopoly（双寡头）**： an oligopoly with two firms)
- Each firm's costs: FC = \$0, ATC = \$10

EXAMPLE: Mobile Phone Duopoly in Smalltown

P	*Q*	Revenue	Cost	Profit
$0	140	$0	$1,400	–1,400
5	130	650	1,300	–650
10	120	1,200	1,200	0
15	110	1,650	1,100	550
20	100	2,000	1,000	1,000
25	90	2,250	900	1,350
30	80	2,400	800	1,600
35	70	2,450	700	1,750
40	60	2,400	600	1,800
45	50	2,250	500	1,750

Competitive outcome:

$P = ATC = MC$

= $10

Q = 120

Profit = $0

Monopoly outcome:

P = $40

Q = 60

Profit = $1,800

EXAMPLE: Mobile Phone Duopoly in Smalltown

- One possible duopoly outcome
  - **Collusion（合谋）**: an agreement among firms in a market about quantities to produce or prices to charge

- China Mobile and China Unicom could agree to each produce half of the monopoly output:
  - For each firm: $Q = 30$, P = \$40, profits = \$900

Collusion vs. Self-interest

P	*Q*
$0	140
5	130
10	120
15	110
20	100
25	90
30	80
35	70
40	60
45	50

Duopoly outcome with collusion:
Each firm agrees to produce $Q = 30$,
earns profit = $900.

Q1: If *China Mobile reneges on the agreement* （违约） and produces $Q = 40$, what happens to the market price? *China Mobile's* profits? Is it in *China Mobile's* interest to renege on the agreement?

Q2: If both firms renege and produce $Q = 40$, determine each firm's profits.

Collusion vs. Self-Interest

- Both firms would have been better off if both stick to the collusion agreement
- If either of them cannot abide by the agreement, one of them is creating more competition against the other
- If neither of them abides by the agreement, they are eroding the profit margin of themselves
- In reality，each firm has opportunistic incentive （机会主义动机） to renege on the agreement
- **Lesson:** It is difficult for oligopoly firms to form collusions and honor their agreements

Collusion vs. Self-Interest

- A and B are the only two firms of a niche market. Their cost structures are $MC_A=2Q_A$, $MC_B=2+2Q_B$. The market demand is $Q=6-2P$

- Now A and B merge into a single firm C. Calculate the optimal output level of firm C.



Types of Collusion

- **Tacit collusion（隐性共谋）/Mutual forbearance（相互克制）**: Firms indirectly coordinate actions by signaling to others their intention to reduce output and maintain pricing above competitive levels.

- **Explicit collusion（公开共谋）**: Firms directly negotiate output and pricing and divide markets in order to reduce competition
  - **Cartel（卡特尔）**: a group of firms acting in a united manner
    - ✓ Setting the same price; reduce production to get a higher profit margin
    - ✓ Usually cartels are prohibited in all countries, e.g. PVC collusion fined in China 2017
    - ✓ Very few cartels have obtained legitimate status, e.g. the International Air Transport Association (IATA) advises uniform prices for transatlantic airlines/ OPEC（石油出口国组织；“欧佩克国家”）: Organization of the Petroleum Exporting Countries

Revisit the Oligopoly Equilibrium

P	***Q***
$0	140
5	130
10	120
15	110
20	100
25	90
30	80
35	70
40	60
45	50

- If each firm produces ***Q*** = 40,
  market quantity = 80
  P = $30
  each firm's profit = $800

- Is it in *China Mobile*'s interest to increase its output further, to ***Q*** = 50?

- Is it in *China Unicom*'s interest to increase its output to ***Q*** = 50?

Measuring Market Concentration

How to determine the market structure?

Measuring Market Concentration

- Concentration ratio（市场集中度）: the percentage of the market's total output supplied by its (usually) top N largest firms.
  - Top 4 largest firms: CR4
  - Top 5 largest firms: CR5

  … …
- Herfindahl index, or Herfindahl-Hirschman Index (HHI, 赫芬达尔——赫希曼指数)
  - the sum of the squares of the market shares of the 50 largest firms (or summed over all the firms if there are fewer than 50.
  - The higher the concentration ratio and HHI, the less competition.

Range of HHI	Market Structure
HHI<0.01	Perfect Competition
0.01<HHI<0.15	Monopolistic Competition
HHI>0.15	Oligopoly or Monopoly

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Concentration Ratios in Chinese Industries

Industry (top 3-5 players)	Ratio
Third-party mobile payment Alipay 54%, WeChatpay 40%,	94%
Beer Snow 31.9%, Tsingtao 22.9%, Budweiser 19.5%, Yanjing 10.3%, Carlsberg 7.4%	92%
Search engines Baidu 85.48%, Bing 4.44%, Sogou 3.66%, Google 2.93%,	95.51%
Smart phone (units) Vivo 19.2%, Apple 18%, Oppo 17.5%, Honor 16.7%, Xiaomi 11%,	85%